

Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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Abstract—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical patterns, using ~ 3.94 million competition results from the public-domain OpenPowerlifting database. Four findings, each answered with a course-appropriate test and led by effect size (at $n \approx 3.9\text{M}$ almost everything is “significant”). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF class limit (de-heaped $\log(\text{below/above}) = +1.92$ at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at all seven limits (surviving Holm and Benjamini–Hochberg), a non-limit control shows no bunching, and placebos show no bunching-direction false positives. (iii) *Allometry*: strength scales with bodyweight differently by sex ($b = 0.75$ men, 0.51 women vs. the isometric $2/3$; a $\text{Sex} \times \log(\text{BW})$ interaction is highly significant and persists on common support). (iv) *Prediction*: bodyweight, sex, equipment, and age predict the total ($R^2 = 0.70$, random forest, leakage-safe grouped CV). These choices leave measurable patterns in the data. Code and data: <https://github.com/adirelm/opl-stat-theory>.

I. INTRODUCTION

Powerlifting comprises three lifts (squat, bench press, deadlift), each with three attempts, contested within bodyweight classes. Two features make it useful for studying strategic behaviour. First, the implements are *discrete*: plates come in fixed denominations, so the load a competitor can place on the bar is not a continuous choice. Second, eligibility is governed by a *threshold*: a lifter who weighs even slightly above a class limit is bumped into a heavier, stronger field. We focus on the two numbers a competitor records: the load chosen for each attempt, and the bodyweight at weigh-in. Both interact with a sharp administrative rule. We ask whether these two choices leave detectable statistical signatures, and what $\sim 3.94\text{M}$ results reveal about the limits of human strength.

Related work. Manipulation of a “running variable” around an administrative threshold produces *bunching*, a standard tool in public economics [1] that is classically tested with the McCrary density-discontinuity test [2] and its modern, bias-corrected local-polynomial form due to Cattaneo, Jansson, and Ma [3]. Rapid pre-weigh-in weight cutting is well documented in weight-class sports. On the physiological side, the square-cube law predicts that strength, governed by muscle cross-sectional area, scales with bodyweight as $\propto \text{BW}^{2/3}$. Extreme-value theory has been used to estimate the ultimate limits of

athletic performance from record data [6]. On OpenPowerlifting specifically, recent work studies *scoring and scaling* systems [7]; to our knowledge, no formal manipulation test of weight-cut bunching has been applied to this dataset.

Contribution. We organize the paper around the two recorded choices, load and weigh-in bodyweight, and use both halves of the course, classical inference and machine learning. Our contributions are: (1) a rounded-likelihood *quantization* GLRT for the plate grid, with a parametric-bootstrap calibration that respects the boundary nature of the null; (2) to our knowledge the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, checked using a non-limit control, placebo cutoffs, de-heaping, and multiple-testing corrections; (3) a sex-interaction *allometry* model that tests the scaling exponent against the isometric $2/3$; and (4) a leakage-safe strength-*prediction* model. Throughout we lead with effect sizes and confidence intervals rather than p -values, as explained next.

Methodological stance. At $n \approx 3.9\text{M}$ the power to detect any non-zero effect is effectively one, so a p -value answers a question we do not care about (“is the effect exactly zero?”) rather than the one we do (“is it large?”). We therefore report effect sizes with 95% confidence intervals as the primary evidence, use p -values only as a secondary screen, and apply multiple-testing corrections to every pre-declared family: all four (Bonferroni, Holm, Šidák, Benjamini–Hochberg) for the H2 limits, and Holm for H3. Because the same lifter appears in many rows, naive standard errors understate uncertainty (pseudo-replication); the formal tests therefore use one row per lifter or cluster-robust standard errors, declared per hypothesis.

II. RESULTS

All intervals are 95%. H1 uses attempt loads; the cross-sectional tests use one row per lifter (to respect independence), and where the class scheme matters the modern kg-only era (2014+). Headline effect sizes use all rows.

A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI [96.19, 96.21]) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides, so 2.5 kg is the minimum increment. To test this formally we restrict to the kg-native (0.5 kg) lattice and model the within-cell position as a mixture that places weight π on the

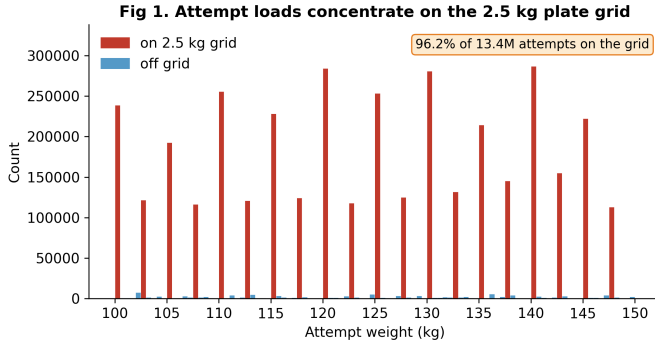


Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid.

grid point and $1 - \pi$ on a uniform background. The GLRT of $\pi = 0$ versus $\pi > 0$ has its null parameter on the boundary of $[0, 1]$, so Wilks' theorem fails: the null distribution of $2 \ln \Lambda$ is the 50:50 mixture $\frac{1}{2} \chi_0^2 + \frac{1}{2} \chi_1^2$ of Chernoff and Self-Liang [4], [5]. Our parametric bootstrap closely matches this (point mass 0.51 at zero; calibrated 95% cutoff 2.59 versus the naive 3.84). The observed statistic, $2 \ln \Lambda = 3.9 \times 10^7$, exceeds every bootstrap draw ($p < 3.3 \times 10^{-4}$), and a χ^2 goodness-of-fit against a uniform within-cell distribution strongly rejects it (5.0×10^7 , $df = 4$). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg = 225 lb), themselves quantized on a different grid.

B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (exact round/half-kilogram weigh-ins removed, so the signal is not mere digit preference), the log-ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ($+1.92 \pm 0.04$, a $6.8 \times$ asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat-to-slightly-negative (-0.21). The formal Cattaneo–Jansson–Ma density-discontinuity test [3] (one random meet per lifter, 2014+) rejects at all seven limits with the density dropping at the cutoff ($t < 0$), surviving both Holm and Benjamini–Hochberg correction; the control shows no excess below (a small *opposite*-direction discontinuity), and no placebo shows the bunching pattern. The effect is robust across equipment (raw $+2.14$, equipped $+1.51$) and tested status (tested $+1.93$), and persists on the clean kg-only subset ($+1.75$). The weakest bunching is at the 120 kg cutoff, which is adjacent to the unbounded 120+ class where the incentive to cut is weakest.

Robustness and falsification. The result survives four checks. (i) *De-heaping*: removing exact round/half-kilogram weigh-ins leaves the asymmetry intact, so the signal is not digit preference and is consistent with weight-cut manipulation. (ii) *Control*: the non-limit point at 91 kg shows no excess below. (iii) *Placebos*: cutoffs placed between real limits show no bunching-direction discontinuity after correction. (iv) *Subgroups and correction*: the effect holds within raw, equipped,

TABLE I
DE-HEAPED $\log(\text{BELOW}/\text{ABOVE})$ AT EACH IPF MEN'S LIMIT (ALL ROWS).
LAST ROW IS A NON-LIMIT CONTROL.

Limit (kg)	log-ratio	$\pm$	$\times$ asymmetry
59	+0.99	0.04	2.7
66	+1.18	0.04	3.2
74	+1.00	0.03	2.7
83	+1.92	0.04	6.8
93	+1.65	0.04	5.2
105	+1.17	0.04	3.2
120	+0.74	0.06	2.1
91 (control)	-0.21	0.03	0.8

and tested subsets, and the formal density test rejects at all seven limits even under the strictest (Bonferroni) correction.

C. Allometric scaling differs by sex (H3)

Fitting $\log(\text{Total}) = a + b \log(\text{BW})$ (Fig. 4), the exponent is $b = 0.75$ (CI $[0.74, 0.75]$) for men and 0.51 ($[0.50, 0.51]$) for women, with heteroskedasticity-robust (HC3) standard errors on one row per lifter. A Wald test rejects the isometric $b = 2/3$ for both sexes (men $z = +46.8$, women $z = -64.1$); the one-sided framing is motivated by Neyman–Pearson/UMP logic, exact only under a one-parameter exponential-family model. A pooled $\text{Sex} \times \log(\text{BW})$ interaction estimates the men-minus-women slope gap at $+0.24$ ($p \ll 10^{-3}$). The gap also *grows* on the common bodyweight support (0.84 men vs. 0.38 women), so it is not an artifact of the sexes occupying different weight ranges. For men, the Pearson correlation of $\log \text{BW}$ and $\log \text{Total}$ is 0.60 (Fisher-transform CI $[0.60, 0.60]$; Spearman 0.58); it is lower for women. The message is not that women are weaker, but that the *estimated scaling* differs; we leave the mechanism to future work.

D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), with every split grouped by lifter and all preprocessing fit inside each training fold, a random forest reaches $R^2 = 0.70$ (RMSE 86 kg) versus 0.61 (RMSE 98 kg) for linear regression (Fig. 5); permutation importance is dominated by sex and bodyweight, and continuous-feature VIFs are ≈ 1 (no collinearity concern). A logistic “made-weight” classifier built from *non-bodyweight* features (age, tested status, era, equipment; the Dots score is excluded because it is a function of total and bodyweight) reaches $\text{AUC} = 0.58$ (PR-AUC 0.88, balanced accuracy 0.54). The signal is weak, as expected: cutting is driven by bodyweight relative to the limit, which defines the label and which these features omit.

E. Supporting analyses

Distribution structure. The apparent two-component shape of raw totals is just sex: by BIC and a bootstrap likelihood-ratio test, raw Total prefers two Gaussian components ($\Delta \text{BIC} = 1.4 \times 10^3$) but the sex-normalized Dots score

Fig 2. Bodyweight bunches just below a real class limit, not at a non-limit control

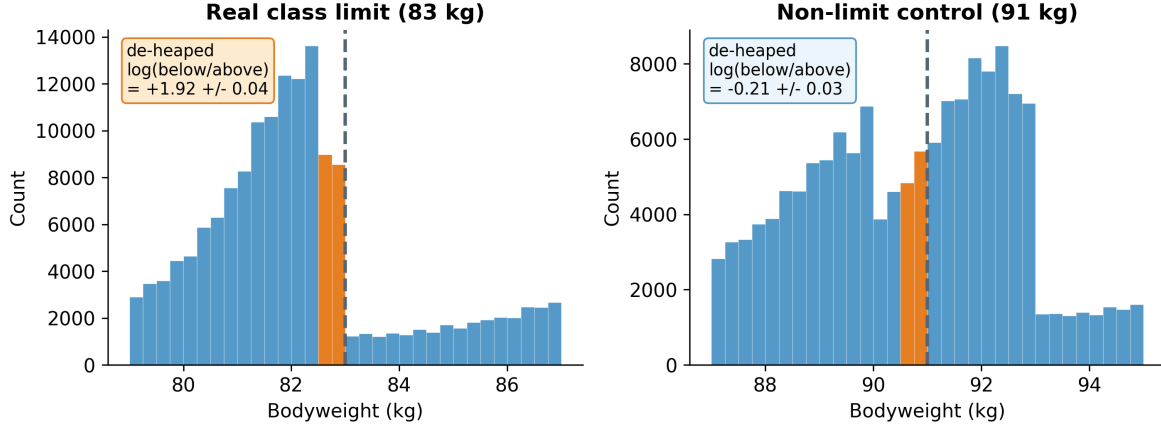


Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

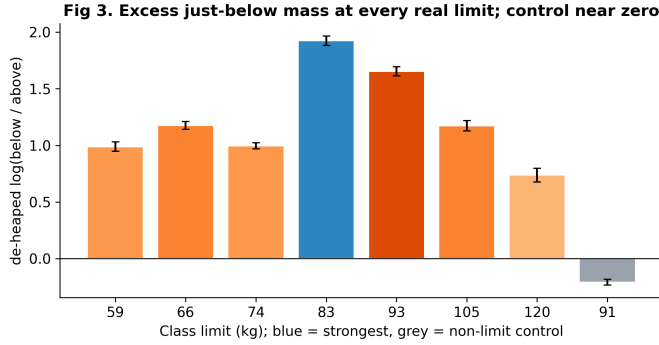


Fig. 3. H2: excess just-below mass at every real limit; control near zero.

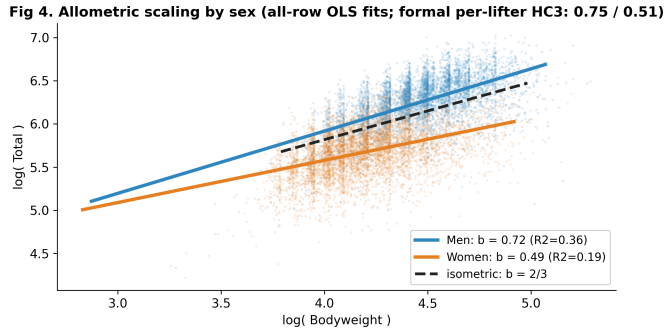


Fig. 4. H3: allometric scaling by sex vs. the isometric 2/3.

prefers one (Fig. 6); the two-component shape largely disappears after Dots normalization. **Tested vs. untested.** Untested competitions show higher Dots than tested ones (medians 344 vs. 318, Mann–Whitney $p \rightarrow 0$, rank-biserial -0.20), consistent with the expected effect of performance-enhancing drugs. **Extreme values.** A GEV fit to equal-size block maxima of recent totals gives shape $\xi = -0.02$ with bootstrap CI $[-0.20, 3.77]$: a population strength *ceiling is not supported*, since the CI includes zero. **Time trend.** Median Dots drifted

down over 2000–2024 (Spearman $\rho = -0.54$), one possible explanation is broadening participation; the trend is not by itself evidence of declining strength. **Sequential detection.** A Wald sequential probability ratio test on the chronological stream of just-below/above indicators accepts the bunching hypothesis after a stopping time of only $N = 27$ (Fig. 8). **Independence.** After deduplicating to one row per lifter, the association between tested status and cutting is negligible (Cramér’s $V = 0.008$), illustrating how pseudo-replication inflates significance. **Residual normality.** The H3 residuals are left-skewed with a heavy lower tail (skew -0.77 ; Fig. 7), so they are not exactly normal. This does not undermine the slope: at this n its sampling distribution is asymptotically normal under standard conditions, and we report HC3-robust standard errors. A formal normality test rejects beyond the smallest samples, as expected at this scale, so we treat it as a diagnostic and lead with effect sizes.

III. METHODS

Data and preprocessing. We use a pinned snapshot of OpenPowerlifting (3,941,811 rows, public domain; the SHA-256 is recorded for exact reproducibility). The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor, so the grid is robust to clustering) and one *row per lifter* for the cross-sectional tests; *de-heaping* drops exact round/half-kilogram weigh-ins so the bunching signal is not digit preference. For prediction, all cross-validation is grouped by lifter so the same athlete never appears in both train and test.

GLRT and the boundary (H1). With log-likelihood ℓ , the statistic is $2 \ln \Lambda = 2(\ell_1 - \ell_0)$. Under regularity $2 \ln \Lambda \rightarrow \chi_r^2$, but when the null fixes a parameter on the boundary of its space the limit becomes a mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [4], [5]; we estimate the null distribution directly by simulating data under the fitted null model (parametric bootstrap) and comparing the observed statistic to the simulated quantiles. Concretely, for

Fig 5. Predicting strength: bodyweight + sex dominate; RF beats linear

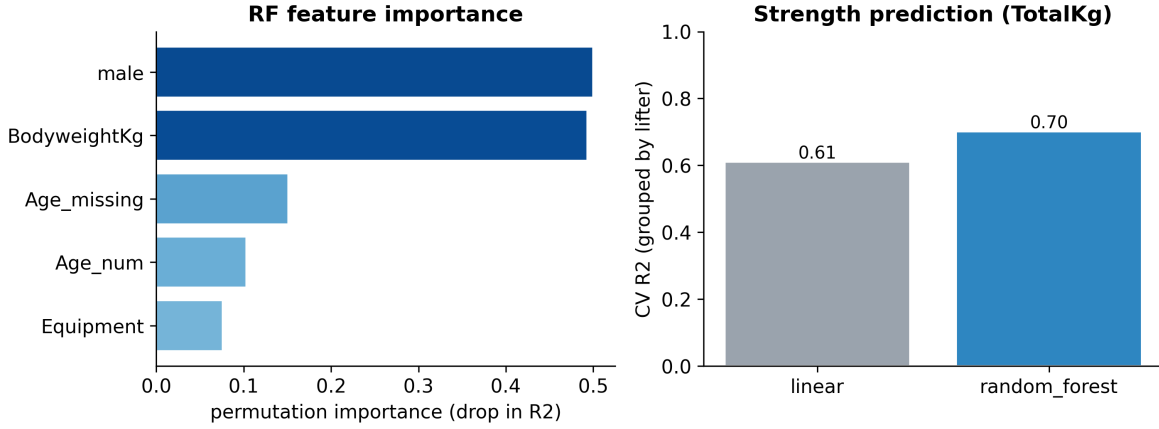


Fig. 5. H4: strength prediction; importance and grouped-CV R^2 .

Fig 6. The apparent 'mixture' in raw Total is just sex; Dots removes it

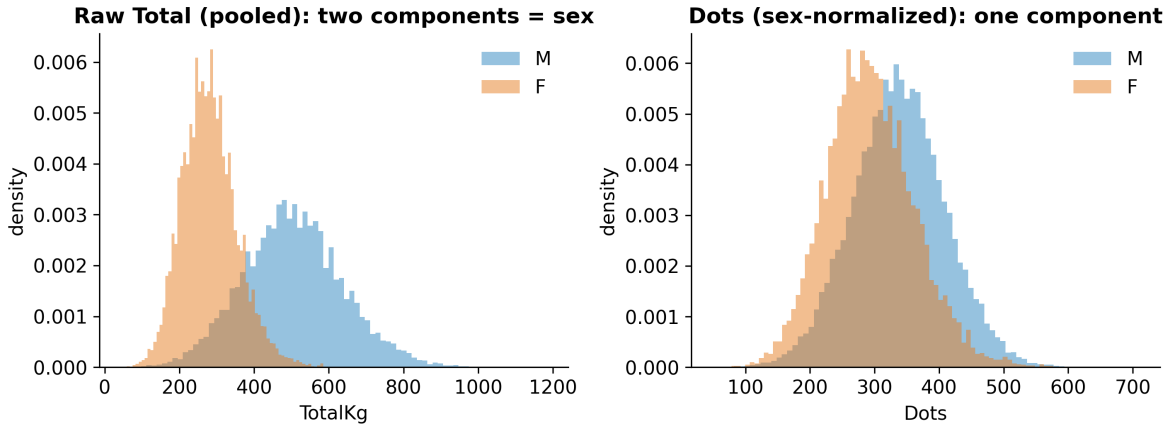


Fig. 6. The apparent “mixture” in raw Total is sex; Dots removes it.

within-cell position $u \in \{0, \dots, 4\}$ on the 0.5 kg lattice the model is

$$P(u) = \pi \mathbb{I}[u = 0] + (1 - \pi) \frac{1}{5}, \quad \pi \in [0, 1], \quad (1)$$

and the GLRT contrasts $\pi = 0$ (no grid preference, on the boundary) with the constrained MLE $\hat{\pi} \geq 0$.

Density discontinuity (H2). For a cutoff c the manipulation test contrasts the one-sided densities, $\theta = \ln \hat{f}^+(c) - \ln \hat{f}^-(c)$, estimated by local polynomials without pre-binning [2], [3]; $\theta < 0$ ($t < 0$) indicates excess mass just below c . We complement the formal test with a de-heaped log count-ratio as the interpretable effect size.

Allometry (H3). OLS on log-log data gives the scaling exponent b ; we report HC3-robust standard errors (one row per lifter), a Wald statistic $W = (\hat{b} - 2/3)/\widehat{\text{se}}(\hat{b})$, and a pooled Sex $\times$ log(BW) interaction whose coefficient is exactly the men-minus-women slope difference. Correlations use the Fisher z -transform for confidence intervals.

Learning and the rest of the toolbox. Prediction uses linear and random-forest regression and logistic classification, evaluated by grouped cross-validation (R^2 , adjusted R^2 , RMSE, MAE, permutation importance, VIF, AUC/PR-AUC). Supporting analyses use a bootstrap mixture-LRT with AIC/BIC, a GEV extreme-value fit, a Wald sequential probability ratio test (with decision boundaries $A = \ln \frac{1-\beta}{\alpha}$, $B = \ln \frac{\beta}{1-\alpha}$, a stopping time, and an expected-sample-size approximation), one- and two-sample nonparametric tests (Wilcoxon signed-rank; Mann-Whitney), an F -test/Kruskal-Wallis omnibus with post-hoc comparisons, and a χ^2 independence test with Cramér’s V and standardized residuals. The extreme-value fit uses the GEV distribution

$$G(x) = \exp \left\{ - \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right\}, \quad (2)$$

where $\xi < 0$ implies a bounded upper tail (a strength ceiling) and $\xi \geq 0$ does not; we estimate ξ from equal-size block maxima and bootstrap its interval. The sequential test’s

Fig 8. H3 residuals: a heavy lower tail (non-normal); the slope is CLT-robust at large n

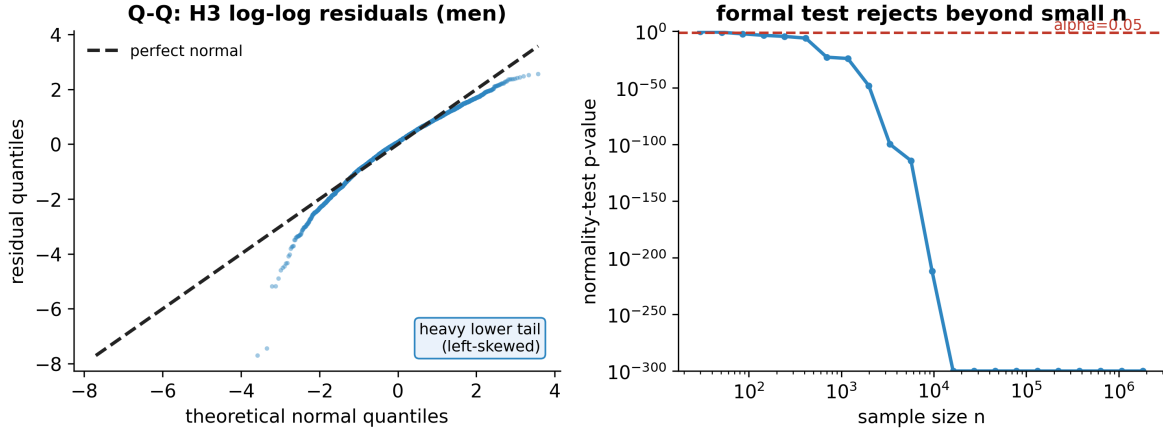


Fig. 7. H3 residuals: heavy lower tail; the slope is CLT-robust at large n .

Fig 7. Sequential test (stopping time) and a-priori power

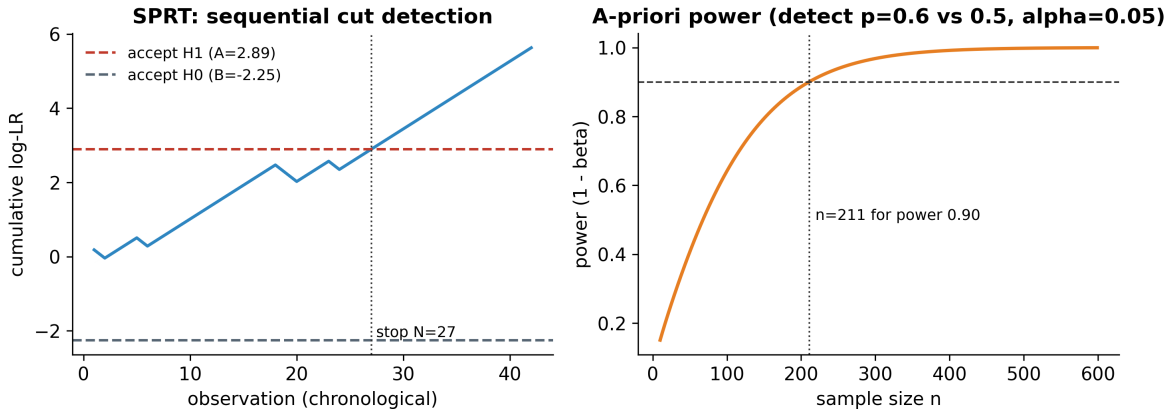


Fig. 8. Sequential test (stopping time $N = 27$) and a-priori power ($n = 211$ for power 0.90).

expected sample size under the alternative is approximated by $\mathbb{E}[N | H_1] \approx [(1 - \beta)A + \beta B] / [p_1 s_1 + (1 - p_1)s_0]$, with s_0, s_1 the per-observation log-likelihood increments. Table II maps the course toolbox onto the study.

Reproducibility. Every result is produced by a small set of scripts from the pinned snapshot; numbers are written to machine-readable files, figures are exported at 300 DPI, and an independent re-derivation from the raw data reproduced every headline value reported here.

IV. DISCUSSION

Conclusions. The two numbers a competitor controls leave measurable, falsification-surviving traces. Attempt loads are quantized to the plate grid; bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Beyond strategy, the data also bear on the limits of strength: scaling with bodyweight is sex-specific and departs from the isometric $2/3$, and basic meet-record variables already explain $\sim 70\%$ of the variance in the total. The

same dataset supports the descriptive, testing, and prediction analyses.

Limitations and null results. This is an observational study: we identify population-level patterns consistent with weight cutting, not the intent of any individual. We report three null results as genuine findings. The unbounded 120+ class shows the weakest bunching, consistent with weaker pressure at that unbounded boundary; the extreme-value analysis does *not* support a strength ceiling (the ξ confidence interval includes zero); and we find no material association between tested status and cutting once pseudo-replication is removed. On the methods side, lifter identity is name-based, the per-lifter meet is sampled at random (we report sensitivity), and the formal bunching test is restricted to the modern kg-only era. Several extensions remain for future work: more federations and eras, a structural bunching model that recovers the share of manipulators, and a multiverse robustness check.

TABLE II
COVERAGE OF THE COURSE TOOLBOX.

Concept	Where used
GLRT, MLE, χ^2 goodness-of-fit	H1 grid model
Boundary LRT, parametric bootstrap	H1 null calibration
Density discontinuity (McCrary/CJM)	H2 bunching
Multiple testing (Bonf./Holm/Šidák/BH)	H2 limits, H3
OLS, R^2 , Wald, interaction term	H3 allometry
One-/two-sided, one-sample tests	H1, H3
Pearson/Spearman, Fisher CI	H3
Regression, random forest, logistic	H4 prediction
Learning metrics, grouped CV, VIF	H4
Mixture LRT, AIC/BIC	structure
Mann–Whitney (two-sample)	tested vs. untested
Wilcoxon signed-rank (paired)	attempt 1 vs. 3
F /ANOVA, Kruskal–Wallis, post-hoc	equipment
χ^2 independence, Cramér’s V	tested $\times$ cut
Extreme-value theory (GEV)	strength ceiling
Sequential Wald (SPRT), stopping time	cut detection
Power, Type I/II tradeoff	a-priori power

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